Maxwell's Equations are a set of four fundamental equations in electromagnetism that describe how electric and magnetic fields interact. In their differential form, they are expressed as:

1. **Gauss's Law for Electricity**:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

This equation states that the electric flux through a closed surface is proportional to the charge enclosed.

2. **Gauss's Law for Magnetism**:

$$\nabla \cdot \mathbf{B} = 0$$

This indicates that there are no magnetic monopoles; magnetic field lines are continuous and have no beginning or end.

3. **Faraday's Law of Induction**:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

This describes how a time-varying magnetic field induces an electric field.

4. **Ampère-Maxwell Law**:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

This relates magnetic fields to the currents and changing electric fields that produce them.

Where:

- $\mathbf{E}$ is the electric field,
- $\mathbf{B}$ is the magnetic field,
- ρ is the electric charge density,
- $\mathbf{J}$ is the current density,
- ε_0 is the permittivity of free space,
- μ_0 is the permeability of free space.